

Problem Set # 14.

S 12.5, 6

$$J_z^{(1)} = k \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\text{Hence } |1,1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$D^{(1)}(\theta_x \hat{x}) = (\cos \theta_x - 1) \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$-i \sin \theta_x \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$D^{(1)}(\theta_x \hat{x}) |1,1\rangle = \begin{pmatrix} 1 + \frac{1}{2} (\cos \theta_x - 1) \\ -\frac{i}{\sqrt{2}} \sin \theta_x \\ \frac{1}{2} (\cos \theta_x - 1) \end{pmatrix}$$

$$\neq |1,0\rangle$$

because $1 + \frac{1}{2} (\cos \theta_x - 1)$ and

$\frac{1}{2} (\cos \theta_x - 1)$ can not simultaneously

vanish.

$$\langle \vec{J} \rangle = \langle 1, 1 | D^{(1)}(\theta_x \hat{x})^\dagger \vec{J} D^{(1)}(\theta_x \hat{x}) | 1, 1 \rangle$$

$$|1, 1\rangle_{\theta_x} \equiv D^{(1)}(\theta_x \hat{x}) |1, 1\rangle = \frac{1}{2} \begin{pmatrix} 1 + \cos \theta_x \\ -i\sqrt{2} \sin \theta_x \\ -1 + \cos \theta_x \end{pmatrix}$$

$${}_{\theta_x} \langle 1, 1 | = \frac{1}{2} (1 + \cos \theta_x, i\sqrt{2} \sin \theta_x, -1 + \cos \theta_x)$$

$$\langle J_x \rangle = {}_{\theta_x} \langle 1, 1 | \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} |1, 1\rangle_{\theta_x}$$

$$= \frac{\hbar}{4\sqrt{2}} \left[(1 + \cos \theta_x)(-i\sqrt{2} \sin \theta_x) + (i\sqrt{2} \sin \theta_x)(1 + \cos \theta_x - 1 + \cos \theta_x) + (-1 + \cos \theta_x)(-i\sqrt{2} \sin \theta_x) \right] = 0$$

$$\langle J_y \rangle = {}_{\theta_x} \langle 1, 1 | \frac{\hbar i}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} |1, 1\rangle_{\theta_x}$$

$$= \frac{\hbar i}{4\sqrt{2}} \left[-(1 + \cos \theta_x)(-i\sqrt{2} \sin \theta_x) + (i\sqrt{2} \sin \theta_x)(1 + \cos \theta_x + 1 - \cos \theta_x) + (-1 + \cos \theta_x)(-i\sqrt{2} \sin \theta_x) \right]$$

$$= \frac{\hbar}{4\sqrt{2}} 4\sqrt{2} \sin\theta_x = -\hbar \sin\theta_x$$

$$\langle J_z \rangle = \langle 1, 1 | \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} | 1, 1 \rangle_{\theta_c}$$

$$= \frac{\hbar}{4} [(1 + \cos\theta_x)^2 - (-1 + \cos\theta_x)^2]$$

$$= \hbar \cos\theta_x$$

$$\langle \vec{J} \rangle = \hbar \left(\hat{x} \cos\theta_x - \hat{y} \sin\theta_x \right)$$

S 12.5.7

$$U(\alpha, \beta, \gamma) \equiv e^{-\frac{i}{\hbar} \alpha J_z} e^{-\frac{i}{\hbar} \beta J_y} e^{-\frac{i}{\hbar} \gamma J_z}$$

i

$$e^{-\frac{i}{\hbar} \beta J_y^{(1)}} = (\cos \beta - 1) \left(\frac{1}{\hbar} J_y^{(1)} \right)^2 - i \sin \beta \left(\frac{1}{\hbar} J_y^{(1)} \right) + 1_{3 \times 3}$$

$$= (\cos \beta - 1) \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix} + \sin \beta \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D^{(1)}(\alpha, \beta, \gamma) = \begin{pmatrix} e^{-i\alpha} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\alpha} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 + \cos \beta & -\sqrt{2} \sin \beta & 1 - \cos \beta \\ \sqrt{2} \sin \beta & 2 \cos \beta & -\sqrt{2} \sin \beta \\ 1 - \cos \beta & \sqrt{2} \sin \beta & 1 + \cos \beta \end{pmatrix} \begin{pmatrix} e^{-i\gamma} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\gamma} \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{-i(\alpha+\gamma)} (1 + \cos \beta) & -\sqrt{2} e^{-i\alpha} \sin \beta & e^{-i(\alpha-\gamma)} (1 - \cos \beta) \\ \sqrt{2} e^{-i\gamma} \sin \beta & 2 \cos \beta & -\sqrt{2} e^{i\gamma} \sin \beta \\ e^{i(\alpha-\gamma)} (1 - \cos \beta) & e^{i\alpha} \sqrt{2} \sin \beta & e^{i(\alpha+\gamma)} (1 + \cos \beta) \end{pmatrix}$$

ii

$$|1, 1\rangle_{\alpha\beta\gamma} \equiv D^{(1)}(\alpha, \beta, \gamma) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{-i(\alpha+\gamma)} (1 + \cos \beta) \\ \sqrt{2} e^{-i\gamma} \sin \beta \\ e^{i(\alpha-\gamma)} (1 - \cos \beta) \end{pmatrix}$$

$$\langle 1, 1 |_{\alpha\beta\gamma} = \frac{1}{2} \left(e^{i(\alpha+\gamma)} (1 + \cos \beta), \sqrt{2} e^{i\gamma} \sin \beta, e^{-i(\alpha-\gamma)} (1 - \cos \beta) \right)$$

$$\langle J_x \rangle = \langle 1, 1 | \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} | 1, 1 \rangle_{\alpha\beta\gamma}$$

$$= \frac{\hbar}{4\sqrt{2}} \left[\sqrt{2} e^{+i\alpha} (1 + \cos\beta) \sin\beta \right. \\ \left. + \sqrt{2} \sin\beta (e^{-i\alpha} (1 + \cos\beta) + e^{+i\alpha} (1 - \cos\beta)) \right. \\ \left. + e^{-i\alpha} (1 - \cos\beta) \sqrt{2} \sin\beta \right]$$

$$= \frac{\hbar}{4} \sin\beta \cdot 4 \cos\alpha = \hbar \sin\beta \cos\alpha$$

$$\langle J_y \rangle = \langle 1, 1 | \frac{\hbar i}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} | 1, 1 \rangle_{\alpha\beta\gamma}$$

$$= \frac{\hbar i}{4\sqrt{2}} \left[-\sqrt{2} e^{+i\alpha} (1 + \cos\beta) \sin\beta + \sqrt{2} \sin\beta (e^{-i\alpha} (1 + \cos\beta) - e^{+i\alpha} (1 - \cos\beta)) \right. \\ \left. + e^{i\alpha} (1 - \cos\beta) \sqrt{2} \sin\beta \right]$$

$$= \frac{\hbar i}{4} \sin\beta (-2 e^{+i\alpha} + 2 e^{-i\alpha}) = + \hbar \sin\beta \sin\alpha$$

$$\langle J_z \rangle = \langle 1, 1 | \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} | 1, 1 \rangle_{\alpha\beta\gamma}$$

$$= \frac{\hbar}{4} (1 + \cos\beta)^2 - \frac{\hbar}{4} (1 - \cos\beta)^2 = \hbar \cos\beta$$

$$\langle \vec{J} \rangle = \hbar (\sin\beta \cos\alpha \hat{x} + \sin\beta \sin\alpha \hat{y} + \cos\beta \hat{z})$$

iii. $|1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$$|1,1\rangle_{\alpha\beta\gamma} = \frac{1}{2} \begin{pmatrix} e^{-i(\alpha+\gamma)} (1+\cos\beta) \\ \sqrt{2} e^{-i\gamma} \sin\beta \\ e^{i(\alpha-\gamma)} (1-\cos\beta) \end{pmatrix}$$

$e^{-i(\alpha+\gamma)} (1+\cos\beta)$ and $e^{i(\alpha-\gamma)} (1-\cos\beta)$

can not vanish simultaneously

iv. $\langle 1, m' | D^{(1)}(\alpha, \beta, \gamma) | 1, m \rangle$

is the matrix given in part i.

Its elements are generally non-zero.

v. $D^{(1)}(0, \pi, 0) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

$$D^{(1)}(0, \pi, 0) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

S12.5.9

$$\int \psi_1^* (L^2 \psi_2) d\Omega$$

$$= \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta \psi_1(\theta, \varphi)^* (-k^2) \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right) \psi_2(\theta, \varphi)$$

$$= \left(\int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta \left[(-k^2) \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right) \psi_2(\theta, \varphi)^* \right] \psi_1(\theta, \varphi) \right)^*$$

$$\int_0^{2\pi} d\varphi \psi_1(\theta, \varphi) \frac{\partial}{\partial\varphi} \psi_1'(\theta, \varphi) = - \int_0^{2\pi} d\varphi \psi_1'(\theta, \varphi) \frac{\partial}{\partial\varphi} \psi_1(\theta, \varphi)$$

because $\left[\psi_1(\theta, \varphi) \psi_1'(\theta, \varphi) \right]_0^{2\pi} = 0$ (ψ and ψ' are periodic)

$$\int_0^\pi \sin\theta d\theta \psi_1(\theta, \varphi) \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} \right) \psi_1'(\theta, \varphi)$$

$$= - \int_0^\pi d\theta \frac{\partial\psi}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} \psi'$$

$$= \int_0^\pi d\theta \psi' \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial\psi}{\partial\theta} \right)$$

$$= \int_0^\pi \sin\theta d\theta \psi' \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial\psi}{\partial\theta}$$

because the surface terms

$$\psi(\theta, \varphi) \sin \theta \left. \frac{\partial \psi'(\theta, \varphi)}{\partial \theta} \right|_0^{\pi} = 0$$

$$\text{and } \psi'(\theta, \varphi) \sin \theta \left. \frac{\partial \psi(\theta, \varphi)}{\partial \theta} \right|_0^{\pi} = 0$$

Indeed for the kinetic energy

$$\int_0^{2\pi} \int_0^{\pi} \sin \theta d\theta \frac{-\hbar^2}{2m} \psi^*(\theta, \varphi) \nabla^2 \psi(\theta, \varphi)$$

to be finite ψ must be finite

as $\theta \rightarrow 0, \pi$.

S 12.5.10

$$L^2 |\alpha \beta\rangle = \alpha |\alpha \beta\rangle$$

$$-\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \psi_{\alpha \beta}(\theta, \phi) = \alpha \psi_{\alpha \beta}(\theta, \phi)$$

$$\frac{\hbar}{i} \frac{\partial}{\partial \phi} \psi_{\alpha \beta}(\theta, \phi) = \beta \psi_{\alpha \beta}(\theta, \phi)$$

$$\beta = m \hbar$$

$$m = 0, \pm 1, \pm 2, \dots$$

$$\psi_{\alpha m}(\theta, \phi) = P_{\alpha}^m(\theta) e^{im\phi}$$

$$\left(\frac{1}{\sin \theta} \frac{d}{d\theta} \sin \theta \frac{d}{d\theta} + \frac{\alpha}{\hbar^2} - \frac{m^2}{\sin^2 \theta} \right) P_{\alpha}^m(\theta) = 0$$

$$u = \cos \theta$$

$$\frac{d}{d\theta} = \frac{du}{d\theta} \frac{d}{du} = -\sqrt{1-u^2} \frac{d}{du}$$

$$\left(\frac{d}{du} (1-u^2) \frac{d}{du} + \frac{\alpha}{\hbar^2} - \frac{m^2}{\sin^2 \theta} \right) P_{\alpha}^m(\theta) = 0$$

$$\left[(1-u^2) \frac{d^2}{du^2} - 2u \frac{d}{du} + \frac{\alpha}{\hbar^2} \right] P_{\alpha}^0(\theta) = 0$$

-2

0

0

0

$$P_{\alpha}^0 = \sum_{n=0}^{\infty} C_n u^n$$

$$\sum_{n=0}^{\infty} c_n \left[n(n-1) (u^{n-2} - u^n) - 2nu^n + \frac{\alpha}{k^2} u^n \right] = 0$$

$$\sum_{n=0}^{\infty} \left(c_{n+2} (n+2)(n+1) - c_n \left[n(n-1) + 2n - \frac{\alpha}{k^2} \right] \right) u^n = 0$$

$$\frac{c_{n+2}}{c_n} = \frac{n(n+1) - \frac{\alpha}{k^2}}{(n+2)(n+1)} \rightarrow 1 \text{ as } n \rightarrow \infty$$

$$\sum_{\substack{n=0 \\ \text{even}}}^{\infty} c_n u^n \sim \sum_{\substack{n=0 \\ \text{even}}}^{\infty} u^n = \frac{1}{1+u} + \frac{1}{1-u}$$

$$\sum_{\substack{n=1 \\ \text{odd}}}^{\infty} c_n u^n \sim \sum_{\substack{n=1 \\ \text{odd}}}^{\infty} u^n = \frac{1}{1-u} - \frac{1}{1+u}$$

The series diverges at $|u| \rightarrow 1$, unless

$$\frac{\alpha}{k^2} = l(l+1) \text{ for some integer } l \\ l=0, 1, 2, \dots$$

$$P_0(u) = \text{constant} \quad \checkmark$$

$$P_1(u) = \text{constant} \times u \quad \checkmark$$

$$P_2(u) = \text{constant} \times \left(1 + \frac{1}{2} (0-6) u^2 \right)$$

$$= \text{constant} (1 - 3u^2) \quad \checkmark$$

12.5.13

$$\psi(\vec{r}) = N(x+y+2z) e^{-\alpha r}$$

$$\frac{i}{-} \quad Y_1^{\pm 1} = \mp \left(\frac{3}{8\pi} \right)^{1/2} \sin \theta e^{\pm i\varphi}$$

$$= \mp \left(\frac{3}{8\pi} \right)^{1/2} \frac{1}{r} (x \pm iy)$$

$$Y_1^0 = \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta = \left(\frac{3}{4\pi} \right)^{1/2} \frac{z}{r}$$

$$\frac{ii}{-} \quad \psi = N \left(\frac{4\pi}{3} \right)^{1/2} r \left[\frac{1}{\sqrt{2}} (Y_1^{-1} - Y_1^1) \right.$$

$$\left. + \frac{i}{\sqrt{2}} (Y_1^1 + Y_1^{-1}) + 2 Y_1^0 \right]$$

$$= N \left(\frac{4\pi}{3} \right)^{1/2} r \left[Y_1^1 \left(\frac{-1+i}{\sqrt{2}} \right) + Y_1^{-1} \left(\frac{1+i}{\sqrt{2}} \right) \right.$$

$$\left. + 2 Y_1^0 \right]$$

$$P(l_z = +\hbar) : P(l_z = 0) : P(l_z = -\hbar)$$

$$= 1 : 4 : 1$$

$$\therefore P(l_z = +\hbar) = P(l_z = -\hbar) = \frac{1}{6}$$

$$P(l_z = 0) = \frac{4}{6} = \frac{2}{3}.$$

SI2.5.14

$$i. \quad \psi = A \frac{1}{r} e^{-r^2/a^2} \\ = A r e^{-r^2/a^2} \sqrt{\frac{4\pi}{3}} Y_0^0$$

$$ii. \quad Y_l^m(\theta, \varphi) = \langle \theta, \varphi | l, m \rangle \\ \rightarrow \langle \theta, \varphi | e^{-i/\hbar \vec{\theta} \cdot \vec{L}} | l, m \rangle \\ = \langle \theta, \varphi | e^{-i/\hbar \theta L_x} | l, m \rangle$$

$$l=1$$

$$L_x^{(l=1)} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$e^{-i/\hbar \theta L_x^{(l=1)}} = (\cos \theta - 1) \left(\frac{L_x^{(l=1)}}{\hbar} \right)^2$$

$$-i \sin \theta \left(\frac{L_x^{(l=1)}}{\hbar} \right) + I_{3 \times 3} \quad (\text{cf exercise 12.5.5})$$

$$e^{-\frac{i}{\hbar} \theta_x L_x} \quad (l=1) = \frac{1}{2} (\cos \theta_x - 1) \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$- \frac{i}{\sqrt{2}} \sin \theta_x \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

In this basis,

$$|1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1,+1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$e^{-\frac{i}{\hbar} \theta_x L_x} \quad (l=1) \quad |1,0\rangle$$

$$= \begin{pmatrix} \frac{1}{2}(1+\cos \theta_x) & -\frac{i}{\sqrt{2}} \sin \theta_x & \frac{1}{2}(\cos \theta_x - 1) \\ -\frac{i}{\sqrt{2}} \sin \theta_x & \cos \theta_x & -\frac{i}{\sqrt{2}} \sin \theta_x \\ \frac{1}{2}(\cos \theta_x - 1) & \frac{i}{\sqrt{2}} \sin \theta_x & \frac{1}{2}(1+\cos \theta_x) \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$= \frac{-i}{\sqrt{2}} \sin \theta_x |1,+1\rangle + \cos \theta_x |1,0\rangle - \frac{i}{\sqrt{2}} \sin \theta_x |1,-1\rangle$$

$$\psi \rightarrow A r e^{-r^2/a^2} \sqrt{\frac{4\pi}{3}} \left(\cos \theta Y_1^0 \right.$$

$$\left. - i \sin \theta \frac{1}{\sqrt{2}} (Y_1^1 + Y_1^{-1}) \right)$$

$$= A e^{-r^2/a^2} (\cos \theta Y_0^0 - \sin \theta Y_1^0)$$